Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2023

Time: 03 Hours

Course: B.Tech.

Code: 100105

Subject: Workshop Manufacturing Practices

Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): [2 x 7 = 14]

(a) The cutting edge of a chisel should be
(i) Hardened (ii) Tempered
(iii) Hardened and Tempered (iv) Case Hardened

(b) The purpose of a riser is to
(i) Deliver molten metal into mould cavity
(ii) Act as a reservoir for molten metal
(iii) Feed the molten metal to the casting in order to compensate for the shrinkage
(iv) Deliver the molten metal from pouring basin to gate

(c) The limitation of electro-chemical machining (ECM) process is
(i) the use of corrosive media as electrolyte makes it difficult to handle
(ii) poor surface finish
(iii) poor accuracy of work piece dimensions because of large tool wear
(iv) there will be thermal damage to the work piece

(d) In forehand welding, the weld is made
(i) from left to right
(ii) from right to left
(iii) first from left to right then from right to left
(iv) either from left to right or from right to left

(e) Which type of hammer is used for extracting the nails out of wood?
(i) Ball peen hammer (ii) Claw peen hammer
(iii) Cross peen hammer (iii) Straight peen hammer

(f) Which of the following processes of moulding is widely used for the manufacturing of bottle caps and automotive dashboards?
(i) Compression moulding (ii) Transfer moulding
(iii) Injection moulding (iv) Jet moulding

(g) Why are the plasticisers added to polymers?
(i) To hold other constituents of plastic together.
(ii) To reduce the cost and enhance the strength and hardness of plastics.
(iii) To improve flexibility and to reduce the temperature and pressure required for moulding of plastics.
(iv) None of the above.

(h) Which material is commonly used in consumer-grade 3D printers?
(i) Metal (ii) Glass (iii) Plastic (iv) Rubber

(i) In forging process, metals are shaped by ____
(i) Impact (ii) Cohesion
(iii) Tension (iv) None of the mentioned

(j) The liquid temperature of the filler metal used in brazing is _____
(i) 150°C (ii) 427°C
(iii) 723°C (iv) 1000°C

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Q.2 (a) What do you understand by plant layout and shop layout? List out the equipment required for a foundry shop and draw the shop layout. [7]

(b) Write a note on need and development of additive manufacturing system [7]

Q.3 (a) Enumerate the safety precautions to be taken in a workshop. [7]

(b) Distinguish between thermoplastic and thermosetting plastic. [7]

Q.4 (a) Explain the meaning of metal forming. Give the classification of metal forming processes. [6]

(b) What is the difference between hot forming and cold forming? Discuss any three metal forming processes with neat sketches. [8]

Q.5 (a) What is a pattern? Name various allowances associated with pattern and why are they provided? [7]

(b) Discuss the different types of defects in casting process [7]

Q.6 (a) Make a sketch of centre lathe and label its main parts. [8]

(b) Explain any three operations performed on a lathe machine with neat sketches. [6]

Q.7 (a) Sketch the quick return motion mechanism used in shaper and explain why it is provided in it. [7]

(b) List out the different types of files. Write the use of any three types of files with neat sketch. [7]

Q.8 (a) Describe the principle of oxyacetylene gas welding with neat sketch. [7]

(b) Distinguish between welding, brazing and soldering. Which process produces the strongest joints and why? [7]

Q.9 Write short notes on the following:- [3½ x4=14]

(a) CNC machine

(b) Power tools

(c) Seasoning of wood

(d) Glass cutting

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